

Inferential Statistics: Knowledge Score vs. Occupation & Income

One-Way ANOVA (Analysis of Variance)

This report examines whether clinical knowledge differs significantly across the 4 specific occupation categories and the 4 monthly income brackets.

1. Occupation Category ANOVA

Comparing Not employed, Private sector, Government sector, and Self-employed.

Occupation Category	N	Mean (Weighted V3)	Std Dev (Weighted)
Not employed	76	2.269	4.248
Private sector	66	0.686	4.657
Government sector	48	3.188	4.472
Self-employed	8	2.353	5.284

- **F-Statistic:** 3.119
- **P-Value:** 0.0272
- **Conclusion:** Statistically Significant ($p < 0.05$). There is a meaningful difference in knowledge based on occupation category.



2. Monthly Income ANOVA

Comparing the four income brackets among those who reported a salary.

Monthly Income	Mean (Weighted	Std Dev
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	(LKR)	N	V3)	(Weighted)
No Income		74	1.826	4.029
< 25,000		5	5.859	4.426
25,000 – 50,000		42	0.343	5.565
51,000 – 100,000		66	2.205	3.889
> 100,000		14	5.149	4.951

- **F-Statistic:** 4.260
- **P-Value:** 0.0025
- **Conclusion: Statistically Significant ($p < 0.05$).** There is a meaningful difference in knowledge based on income level.



Violin Plot